Nishka Katoch

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AI Engineer with an M.Sc. degree from Mila (Canadian Work Permit), experience building and evaluating computer vision and generative AI systems, with recent research in LLM and agent evaluation.

Education

University of Montreal/Mila — Montreal
Master of Science in Computer Science, Thesis in AI — *Sept. 2021 – Aug. 2025*
Supervised by Prof. Julien Cohen-Adad and Prof. Guy Wolf — *NeuroPoly Lab*

Banasthali University — Jaipur
Bachelor of Technology, Computer Science — *July 2017 – May 2021*

Technical Skills

ML & Research: PyTorch, JAX, Hugging Face Transformers, TensorFlow, CUDA, Weights & Biases
Deployment: TensorRT, ONNX, Docker, Linux, Milvus
Languages & Tools: Python, C/C++, SQL, Git

Experience

AI Engineer (Computer Vision) — Jan. 2026 – Present
Inferigence Quotient — *Bangalore*
- Developed a system to match satellite images to UAV imaging and determine location without GPS, achieving 92% Recall@1 using **Vision Transformers** and IVF retrieval.
- Extended the localization system to visible and thermal imagery for matching in low-visibility conditions.
- Accelerated the scene matching time by 15×, staying within 1% baseline accuracy, so it can run in real time onboard the embedded system, using model quantization and **TensorRT FP16**.

AI Intern/Intern Team Lead — Aug. 2025 – Oct. 2025
Maker's Lab, Tech Mahindra — *Hyderabad*
- Led a team of interns and developed methods to benchmark and improve embedding and retrieval systems for **enterprise-scale RAG**, focusing on latency and accuracy.
- Built a deepfake detector combining **Vision Transformers** and quantum learning.

Graduate Researcher — Jan. 2022 – Aug. 2025
NeuroPoly Lab, Polytechnique Montreal/Mila — *Montreal*
- Improved microscopy segmentation models on unseen data by 12% by evaluating different domain-adaptation methods published at **MICCAI 2024.**
- Generated synthetic histology images with a **latent-diffusion** pipeline, achieving 85% Dice without manual labeling and improving segmentation on unseen data.
- Managed the annotation, dataset, and documentation workflows for **2,000+** multimodal histology samples, coordinating directly with medical professionals using **AxonDeepSeg** in practice.

Earlier Roles — 2018 – 2024
ASR, web, and robotics internships
- Evaluated Hindi speech-recognition models through a literature review and experimental testing, achieving 91% accuracy across five regional dialects.
- Built a computer-vision pipeline using SAM and Stable Diffusion for furniture segmentation and design generation, processing 500+ images.
- Optimized the vision pipeline from 45 to 8 seconds per image through model optimization, batching, and quantization.
- Developed a ROS-Unity UAV swarm-control system with 50 ms latency.

Research & Open Source

Constitutional Alignment under Resource Constraints | *Independent Research* — Feb. 2026 – Apr. 2026
- Investigated LLM agent behavior under different oversight and resource constraints, observing increased **reward hacking** under survival pressure and identifying a related reasoning failure mode.

Hugging Face | *Open Source Contribution* — July 2026
- Reproduced three papers through the **ICML 2026 Agent Reproduction Challenge**, including *Who Can We Trust? LLM-as-a-Jury*, *When Does Predictive Inverse Dynamics Outperform Behavior Cloning?*, and *Uncertainty-Aware Cross-View Geo-Localization.*

ARENA AI-Safety Education | *Course* — Ongoing
- Built working implementations of **alignment methods (RLHF, DPO)**, **mechanistic interpretability** techniques, and model evaluations.

Publications

Unpaired Modality Translation for Pseudo-Labeling [pdf] — MICCAI 2024
*Arthur Boschet, Armand Collin, **Nishka Katoch**, Julien Cohen-Adad*

Multi-Contrast Axon Segmentation via Latent Diffusion Models [pdf] — Thesis 2025
***Nishka Katoch**, Julien Cohen-Adad, Guy Wolf*